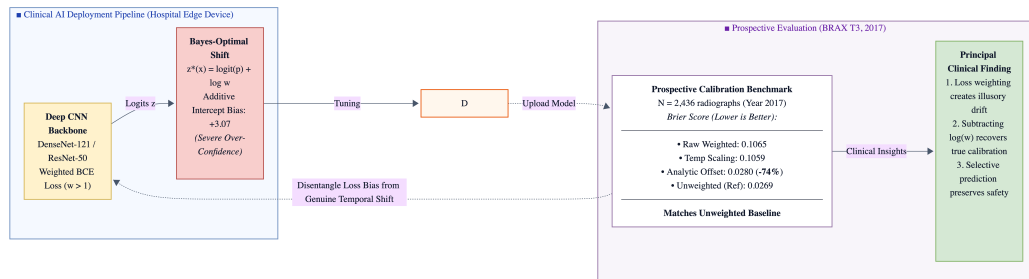


Graphical Abstract

Positive-Class Weighting Induces Systematic Probability Distortion in Deep Learning Models for Chest Radiography under Deidentified-Date Cohort Shift

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Highlights

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- Positive loss weighting induces an additive population logit offset of $\log w$.
- Finite neural network logits empirically cluster tightly around theoretical $\log w$.
- Temperature scaling cannot correct this intercept-type calibration bias.
- Analytic offset ($z - \log w$) cuts 74% of Brier error without retraining.
- Selective triage exposes trade-offs between calibrated risk and screening referral.

Positive-Class Weighting Induces Systematic Probability Distortion in Deep Learning Models for Chest Radiography under Deidentified-Date Cohort Shift

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Abstract

Clinical deep learning models frequently exhibit calibration decay under deidentified-date cohort shift. Under extreme class imbalance, training with positive-weighted binary cross-entropy (BCE) is standard practice, but its downstream effect on probability calibration has been widely conflated with intrinsic feature drift. We resolve this conflation through a controlled $2 \times 2 \times 3$ factorial experiment across 12 models (DenseNet-121 and ResNet-50 trained under unweighted and positive-weighted BCE across 3 seeds) on 40,523 chest radiographs from BRAX. Models are trained on an anchor partition and evaluated across later deidentified-date strata (T_1, T_2, T_3) with strict patient isolation. We show that positive loss weighting mathematically induces an additive population logit shift of $\log w$, which temperature scaling structurally cannot correct. In finite networks, empirical logit differences cluster tightly around theoretical $\log w$ (mean $\Delta z = 3.29 \pm 0.81$ vs. theoretical 3.07 for Pleural Effusion). On stratum T_3 , zero-parameter analytic offset ($z - \log w$) reduces total Brier score by 74% ($\Delta \text{Brier} = -0.0795$ [95% CI: $-0.0893, -0.0697$]), substantially reducing intercept miscalibration from $\alpha = -2.96$ to -0.45 without retraining or validation data. On an external MIMIC-CXR-JPG cohort (3,403 radiographs, 289 patients), discrimination proved pathology-dependent (Pleural Effusion AUROC 0.688; Cardiomegaly 0.523), with severe baseline calibration shift ($\text{BSS} < 0$). Analytic offset yielded significant Brier reductions for Cardiomegaly and Pneumonia, but worsened

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Edema, delineating boundaries of zero-parameter recalibration under cross-institutional shift. Finally, selective prediction simulations expose operational trade-offs between risk estimation and high-sensitivity screening triage.

Keywords: Chest radiography, Probability calibration, Class imbalance, Distribution shift, Selective prediction, Deep learning, Clinical decision support

1. Introduction

Deep learning models for automated chest radiograph interpretation have achieved high discriminatory performance across common thoracic abnormalities such as pleural effusion, cardiomegaly, and pneumonia [1, 2]. However, the transition from stationary retrospective validation to longitudinal clinical deployment introduces substantial distribution shifts. In operational healthcare environments, patient populations evolve, imaging protocols shift, scanner hardware is replaced, and demographic or seasonal patterns alter condition prevalence over time [3, 4, 5].

While discriminatory ranking (measured via AUROC) often exhibits gradual longitudinal attenuation under temporal shift, decision-critical applications rely fundamentally on *calibrated posterior probabilities*: $P(Y = 1 \mid X = x)$ [6, 7]. When predicted probabilities deviate from true empirical event rates, downstream operational decision cutoffs shift unpredictably, expected utility formulations degrade, and automated decision-support pipelines generate distorted risk rankings.

Recent literature has reported rapid “temporal calibration decay,” asserting that frozen deep neural networks suffer progressive miscalibration over deployment periods [8, 9]. Simultaneously, because biomedical datasets are characterized by extreme class imbalance (pathology prevalence often below 5%), researchers routinely train vision models using positive-weighted binary cross-entropy (BCE) to penalize missed positive cases [1, 10].

In this work, we present a computational investigation showing that these two phenomena are deeply entangled, and that the prevailing attribution of temporal calibration degradation primarily to intrinsic feature drift may be largely explained by uncorrected post-training calibration offsets under label distribution shifts [11, 12]. Under population-risk minimization, when models are trained with positive loss weighting $w > 1$, the Bayes-optimal logit is shifted by an additive constant $+\log w$ [13]. Standard post-hoc

calibration methods—most notably temperature scaling [14]—only rescale the logit variance (z/T) and cannot shift the logit intercept. Consequently, weighted models continuously over-predict positive probabilities, producing severe Brier score and Expected Calibration Error (ECE) degradation that compounds over time as disease prevalence fluctuates.

1.1. Related Work

Probability Calibration in Deep Learning: Modern deep convolutional neural networks frequently exhibit over-confidence due to unconstrained cross-entropy minimization with over-parameterized architectures [14]. Post-hoc recalibration techniques such as temperature scaling, histogram binning, and isotonic regression have been widely evaluated [14, 15]. In medical decision models, calibration accuracy is paramount for valid risk stratification [6, 7]. However, standard methods were primarily derived for multi-class classification under balanced distributions, leaving binary imbalanced settings less formally explored.

Class Imbalance and Loss Function Design: Extreme class imbalance is pervasive in thoracic imaging. Common mitigation strategies include positive loss weighting [10, 1], focal loss [16], and re-sampling. While positive weighting successfully shifts operational sensitivity, King and Zeng [13] proved in classical logistic regression that sample proportion adjustments shift the model intercept by $\log w$. Modern deep learning pipelines routinely apply loss weighting without correcting this induced logit offset prior to post-hoc calibration.

Distribution Shift and Label Shift Adaptation: Temporal dataset shift encompasses changes in feature distributions $P(X)$, conditional distributions $P(Y | X)$, and prior label probabilities $P(Y)$ [3]. Davis et al. [12] demonstrated that calibration drift often precedes discriminatory loss in clinical models. Alexandari et al. [11] established that maximum likelihood with bias-corrected calibration provides robust adaptation under label shift. Decoupling algorithmic training offsets from genuine temporal shift is essential for diagnostic integrity.

Selective Prediction and Deferral Mechanisms: Selective classification allows models to abstain from predicting on ambiguous inputs, routing them to human experts [17, 18]. Fisch et al. [19] highlighted that selective classification performance is fundamentally bounded by probability calibration. Investigating how post-hoc calibration interacts with uncertainty-based triage under temporal shift addresses an open operational question.

67 *1.2. Contributions*

68 To disentangle algorithmic loss bias from distribution shift across cohort
69 strata, this paper provides:

- 70 1. **A Controlled Factorial Matrix:** We execute a $2 \times 2 \times 3$ factorial
71 design across 12 models (DenseNet-121 and ResNet-50 architectures $\times$
72 standard BCE and positive-weighted BCE $\times$ 3 random seeds) trained on
73 25,123 chest radiographs and evaluated on 8,302 held-out radiographs
74 across three sequential deidentified-date strata (T_1, T_2, T_3) from the
75 BRAX dataset [2].
- 76 2. **Theoretical Derivation and Empirical Verification of Logit
77 Shift:** Under population risk minimization, we derive that positive-
78 weighted BCE introduces a constant logit offset $\log w$. In finite neural
79 networks, we verify empirically that observed logit differences $(z_w - z_u)$
80 tightly mirror theoretical $\log w$ across multiple thoracic findings. We
81 show that temperature scaling cannot adjust the logit intercept, leaving
82 the calibration intercept severely biased ($\alpha \approx -2.96$), whereas an
83 exact zero-parameter analytic offset ($z - \log w$) or non-negative Platt
84 scaling restores calibration parity with unweighted models (substantially
85 reducing intercept bias to $\alpha = -0.45$ for DenseNet-121 and reducing
86 total Brier error by 74%).
- 87 3. **A Leak-Free Out-of-Sample Calibration Protocol:** We implement
88 a strict 5-fold patient-clustered cross-fitting protocol on T_1 , strictly
89 preventing in-sample calibration leakage, and report out-of-sample gen-
90 eralization on T_2 and T_3 backed by 2,000-replicate patient-clustered
91 bootstrap confidence intervals directly on individual seeds and the 3-seed
92 ensemble.
- 93 4. **Human-in-the-Loop Selective Prediction Simulation:** We for-
94 mulate a selective triage simulation with explicit, transparent tracking
95 of reader referral workloads, retained sensitivity $(TP/N_{\text{pos, ret}})$, and
96 full-cohort system sensitivity $(1 - FN/N_{\text{pos, total}})$ under matched de-
97 velopment thresholds, demonstrating that calibrated models optimize
98 probability estimation while raw weighted models support recall-oriented
99 screening triage.

100 2. Materials and Methods

101 2.1. Dataset and Cohort Stratification

102 We utilize the Brazilian Chest X-ray (BRAX) v1.1.0 benchmark [2]. BRAX
103 v1.1.0 contains 40,967 radiographs. After excluding 444 radiographs from
104 89 patients whose examinations crossed partition boundaries, the analytic
105 cohort comprised 40,523 radiographs acquired across multiple inpatient and
106 outpatient emergency facilities at Hospital Israelita Albert Einstein in São
107 Paulo, Brazil. Radiographs are labeled with 14 thoracic findings extracted
108 from Portuguese radiological reports using a validated natural language
109 processing pipeline.

110 In compliance with HIPAA de-identification standards, direct patient
111 identifiers and original timestamps are protected. We extract chronological
112 study dates from the released 8-digit calendar integers (YYYYMMDD), spanning
113 2008 through 2017. BRAX documentation states that the released dates
114 are fictitious dates generated through a hashing procedure while retaining
115 the original intervals between study acquisitions. Because synchronization
116 of anonymized dates across different patients is not explicitly established,
117 we interpret these partitions as sequential deidentified-date strata rather
118 than real-world calendar periods. To maintain strict partition integrity and
119 preclude longitudinal data leakage:

- 120 1. All studies are partitioned chronologically around a strict model freeze
121 boundary: $T_{\text{freeze}} = 2015-07-01$.
- 122 2. All training studies occur before 2013-09-01 ($N = 25,123$ images from
123 11,265 patients across 14,315 studies).
- 124 3. All validation studies occur between 2013-09-01 and 2015-06-30 ($N =$
125 7,098 images from 3,281 patients across 4,012 studies).
- 126 4. All held-out test studies occur between 2015-07-01 and 2017-12-17
127 ($N = 8,302$ images from 3,807 patients across 4,670 studies), partitioned
128 into three consecutive deidentified-date strata: T_1 (2015-H2, $N = 1,879$),
129 T_2 (2016, $N = 3,987$), and T_3 (2017, $N = 2,436$).
- 130 5. Strict patient isolation is enforced: $\text{Patients}_{\text{train}} \cap \text{Patients}_{\text{val}} \cap \text{Patients}_{\text{test}} =$
131 $\emptyset$. Specifically, 444 radiographs from 89 patients whose examinations
132 crossed partition boundaries were excluded to guarantee zero within-
133 patient contamination, yielding the analytic cohort of 40,523 radio-
134 graphs.

Table 1: Cohort Demographics, Radiographic Projections, and Target Condition Prevalence across Stratified Cohorts

Cohort Stratum	Images (N)	Patients (N)	Studies (N)	Sex (% M / F)	Age (Mean $\pm$ SD)	Views (% PA / AP / Lat / Unk)	Pleural Effusion N (%)	Cardiomegaly N (%)	Pneumonia N (%)	Edema N (%)
Train (Pre-deployment)	25,123	11,265	14,315	49.3% / 50.7%	46.0 $\pm$ 24.3	34.0% / 13.5% / 0.0% / 27.2%	1,116 (4.44%)	2,398 (9.55%)	456 (1.82%)	23 (0.09%)
Val (Pre-deployment)	7,098	3,281	4,012	49.8% / 50.2%	46.2 $\pm$ 24.1	34.7% / 12.2% / 0.0% / 27.7%	295 (4.16%)	681 (9.59%)	128 (1.80%)	8 (0.11%)
Test T1 (2015)	1,879	913	1,039	47.8% / 52.2%	44.5 $\pm$ 23.9	34.4% / 10.2% / 0.0% / 29.0%	99 (5.27%)	147 (7.82%)	50 (2.66%)	4 (0.21%)
Test T2 (2016)	3,987	1,832	2,249	48.5% / 51.5%	46.1 $\pm$ 23.8	34.6% / 12.0% / 0.0% / 27.6%	177 (4.44%)	422 (10.58%)	79 (1.98%)	13 (0.33%)
Test T3 (2017)	2,436	1,164	1,382	50.9% / 49.1%	46.0 $\pm$ 24.6	34.4% / 12.4% / 0.0% / 28.1%	85 (3.49%)	258 (10.59%)	50 (2.05%)	0 (0.00%)
Test All (Pooled)	8,302	3,807	4,670	49.0% / 51.0%	45.7 $\pm$ 24.1	34.5% / 11.7% / 0.0% / 28.1%	361 (4.35%)	827 (9.96%)	179 (2.16%)	17 (0.20%)

6. For longitudinal paired logit offset and multi-calibrator evaluation, all 8,302 held-out radiographs ($N = 1,879$ in T_1 , $N = 3,987$ in T_2 , and $N = 2,436$ in T_3) are evaluated across both clinical architectures (DenseNet-121 and ResNet-50) with zero held-out exclusions (0% exclusion rate), yielding 16,604 paired model evaluations ($8,302 \times 2$) with complete longitudinal tracking.

Demographic, projection, and disease prevalence characteristics across all cohort strata are detailed in Table 1.

2.2. External Transport Validation Cohort (MIMIC-CXR-JPG)

To evaluate cross-institutional transportability under realistic clinical domain shift, frozen BRAX-trained models were externally evaluated on all eligible frontal radiographs (AP and PA views) from the official patient-partitioned MIMIC-CXR-JPG test partition [20, 21]. The cohort comprises $N = 3,403$ frontal radiographs ($N = 3,041$ diagnostic studies across $N = 289$ unique patients; AP projections: 2,405 [70.67%], PA projections: 998 [29.33%]). In strict accordance with the primary evaluation protocol, ground-truth diagnostic labels were extracted using the official CheXpert labeler under the standard uncertainty-zero (U -zero) policy. Empirical condition prevalence across images was: Pleural Effusion 32.18% ($N = 1,095$ positive radiographs), Cardiomegaly 26.33% ($N = 896$), Edema 21.33% ($N = 726$), and Pneumonia 10.05% ($N = 342$). Because multiple radiographs and studies belong to the same patient, patient-clustered non-parametric bootstrap inference was used throughout external evaluation to account for repeated within-patient observations, avoiding artificial inflation of effective sample size. All 3,403 radiographs were physically verified on disk (100.00% completeness). No retraining or hyperparameter fine-tuning was performed, isolating out-of-distribution transportability from in-domain adaptation.

2.3. Architectures and Training Objectives

We evaluate two standard clinical vision architectures initialized from ImageNet pre-trained weights:

- **DenseNet-121** [22]: Employs dense feature concatenation, standard for thoracic pathology benchmarks [1].
- **ResNet-50** [23]: Employs residual skip connections as an independent structural comparator.

Each architecture is trained under two distinct loss objectives:

1. **Unweighted Binary Cross-Entropy (Standard BCE):**

$$\mathcal{L}_{\text{std}}(y, \hat{p}) = -[y \log \hat{p} + (1 - y) \log(1 - \hat{p})] \quad (1)$$

where $\hat{p} = \sigma(z) = 1/(1 + e^{-z})$ is the sigmoid probability from raw logit z .

2. **Positive-Weighted Binary Cross-Entropy (Weighted BCE):**

$$\mathcal{L}_{\text{pos}}(y, \hat{p}) = -[w \cdot y \log \hat{p} + (1 - y) \log(1 - \hat{p})] \quad (2)$$

where $w = (N_{\text{total}} - N_{\text{pos}})/N_{\text{pos}}$ represents the inverse positive prevalence in the training set (e.g., $w = 21.51$ for Pleural Effusion, $w = 9.48$ for Cardiomegaly, $w = 54.09$ for Pneumonia).

Models are trained with the AdamW optimizer (initial learning rate 10^{-4} , weight decay 10^{-2}) and a Cosine Annealing learning rate schedule over 15 epochs, with batch size 64 and Automatic Mixed Precision (AMP fp16) on 512×512 pixel input resolution. Checkpoints are selected based on the epoch achieving the highest validation Pleural Effusion AUROC across the 15 epochs (occurring between epochs 4 and 15 across runs), without early stopping loop termination. Each configuration is executed across three random seeds ($S \in \{42, 123, 2026\}$), producing 12 fully independent models. For unified system-level benchmarking, a 3-seed probability ensemble is constructed for each architecture and loss objective by averaging predicted probabilities across independent training runs:

$$\bar{p}_{\text{raw}}(x) = \frac{1}{3} \sum_{s \in S} \sigma(z_s(x)), \quad \bar{p}_{\text{corrected}}(x) = \frac{1}{3} \sum_{s \in S} \sigma(z_s(x) - \log w) \quad (3)$$

All post-hoc calibration, paired analytic adjustments, and selective prediction policies are evaluated directly on both individual model seeds and the unified probability ensemble, ensuring strict mathematical consistency across all reported metrics and tables.

192 *2.4. Mathematical Mechanics of Loss Weighting and Calibration*

193 Under positive-weighted binary cross-entropy, the empirical risk minimizer
194 seeks to minimize the expected loss:

$$\begin{aligned}\mathbb{E}_{Y|X}[\mathcal{L}_{\text{pos}}(Y, \hat{p})] &= -w \cdot P(Y = 1 | X) \log \hat{p} \\ &\quad - P(Y = 0 | X) \log(1 - \hat{p})\end{aligned}\tag{4}$$

195 Setting the derivative with respect to logit z to zero:

$$\begin{aligned}\frac{\partial}{\partial z} \mathbb{E}[\mathcal{L}_{\text{pos}}] &= \hat{p} [wP(Y = 1|X) + P(Y = 0|X)] \\ &\quad - wP(Y = 1|X) = 0\end{aligned}\tag{5}$$

196 Solving for the optimal predicted probability $\hat{p}^*$:

$$\hat{p}^* = \frac{w \cdot P(Y = 1 | X)}{w \cdot P(Y = 1 | X) + P(Y = 0 | X)}\tag{6}$$

197 Converting to log-odds:

$$z^*(X) = \log \left(\frac{\hat{p}^*}{1 - \hat{p}^*} \right) = \log \left(\frac{P(Y = 1 | X)}{P(Y = 0 | X)} \right) + \log w\tag{7}$$

198 Equation (7) derives that positive loss weighting shifts the model’s population
199 Bayes-optimal logit by $+\log w$. When $w \gg 1$, the network is mathematically
200 incentivized to output large positive logits, systematically inflating predicted
201 probabilities. We emphasize that Equation (7) describes the population-risk
202 minimizer under infinite capacity; in finite neural networks trained with
203 stochastic gradient descent, empirical logits fluctuate around this theoretical
204 attractor.

205 *2.4.1. Limitation of Temperature Scaling*

206 Temperature scaling [14] adjusts probabilities via a single positive scalar
207 T :

$$p_{\text{temp}} = \sigma \left(\frac{z}{T} \right)\tag{8}$$

208 To formally evaluate calibration structure, we examine the standard logistic
209 calibration regression (Cox calibration) [6, 7]:

$$\text{logit}(P(Y = 1 | X)) = \alpha + \beta \cdot \text{logit}(\hat{p})\tag{9}$$

where $\alpha = 0$ represents perfect calibration-in-the-large (calibration intercept), and $\beta = 1$ denotes ideal calibration slope. Because positive loss weighting induces an additive shift $z \approx z_{\text{true}} + \log w$, the uncorrected logit suffers a massive intercept discrepancy ($\alpha \approx -\log w \ll 0$). Temperature scaling multiplies the logit by $1/T$, adjusting the slope $\beta \rightarrow \beta \cdot T$, but is mathematically incapable of shifting the additive constant α . In low-prevalence clinical domains where negative cases vastly dominate, the negative log-likelihood optimizer is constrained by the conflicting pressures of slope and intercept, typically selecting $T^* \approx 1.0$ or providing limited, architecture-dependent variance shrinkage without eliminating the severe underlying intercept displacement.

2.4.2. Post-Hoc Recalibration Suite

To systematically rectify this mechanism, we evaluate four post-hoc recalibration functions:

1. **Temperature Scaling (T^*):** $p = \sigma(z/T^*)$, optimized via bounded Brent minimization on $[0.01, 50.0]$ with an explicit monotonicity constraint (if $\text{NLL}(T^*) > \text{NLL}(1.0) \implies T^* = 1.0$).
2. **Analytic Offset ($z - \log w$):** Directly subtracts the theoretical training logit shift:

$$p_{\text{analytic}} = \sigma(z - \log w) \quad (10)$$

Requires zero validation data and zero numerical optimization, resetting $\alpha \approx 0$ by construction while preserving model discrimination and slope.

3. **Fitted Offset ($z + a^*$):** Optimizes a single scalar intercept $a^* \in [-10.0, 10.0]$ via univariate Brent minimization on validation log-likelihood, holding the slope fixed at 1.0.
4. **Platt Scaling ($a + bz$):** Logistic regression on logits $p = \sigma(a + bz)$ with a non-negative constraint on slope ($b \geq 0$) to prevent spurious rank reversal.

2.4.3. Algorithmic and Computational Complexity

The evaluated post-hoc calibrators exhibit distinct computational profiles:

- **Temperature Scaling:** Incurs $O(N_{\text{val}})$ scalar optimization over validation logits and $O(1)$ inference per prediction.
- **Analytic Offset:** Incurs strictly $O(1)$ closed-form scalar subtraction ($z - \log w$), requiring zero validation storage, zero parameter fitting, and $O(1)$ inference.

- 243 • **Fitted Offset:** Incurs $O(N_{\text{val}})$ univariate Brent optimization on valida-
244 tion log-likelihood and $O(1)$ inference.
- 245 • **Platt Scaling:** Incurs $O(k \cdot N_{\text{val}})$ iteratively reweighted least squares
246 fitting (where k is the number of optimization steps) and $O(1)$ inference.

247 The analytic offset thus provides an operational advantage for zero-compute
248 edge deployment, eliminating the requirement to retain or access historical
249 validation datasets.

250 2.5. Cross-Fitting and Resampling Evaluation Protocol

251 To ensure zero optimistic in-sample bias:

- 252 • On deployment stratum T_1 (2015-H2), calibrator parameters are esti-
253 mated using **5-fold patient-clustered cross-fitting**. Calibrators are
254 fit on 4 folds and evaluated out-of-fold on the 5th fold, with all studies
255 from any given patient assigned to a single fold.
- 256 • For evaluation on later held-out deidentified-date strata T_2 (2016) and
257 T_3 (2017), the calibrators fit on T_1 are frozen and applied strictly
258 out-of-sample.
- 259 • **Statistical Inference:** All 95% confidence intervals (CIs) are com-
260 puted using $B = 2,000$ patient-clustered bootstrap resamples with
261 replacement. For paired calibrator comparisons (e.g., Analytic Offset
262 vs. Raw), differences are computed on identical bootstrap resamples.

263 2.6. Selective Prediction and Deferral Policy

264 In operational deployment, models need not provide an autonomous pre-
265 diction for every image. An uncertain model can selectively defer ambiguous
266 examinations to expert human readers [17, 18].

267 To ensure fair comparisons across calibrators operating on different proba-
268 bility scales, we determine an operating decision cutoff t^* for each method
269 strictly on development stratum T_1 using Youden’s J index ($J = \text{Sensitivity} +$
270 $\text{Specificity} - 1$), without using later held-out evaluation labels. For unweighted
271 models, $t^* \approx 0.014$; for analytic offset calibrated models, $t^* \approx 0.022$; and for
272 raw weighted models, $t^* \approx 0.33$. We define a threshold-relative uncertainty
273 metric based on piecewise distance from the operational cutoff t^* :

$$u(x) = 1 - \frac{|\hat{p}(x) - t^*|}{\mathbb{I}(\hat{p}(x) \geq t^*)(1 - t^*) + \mathbb{I}(\hat{p}(x) < t^*)t^*} \in [0, 1] \quad (11)$$

where $u(x) \rightarrow 1$ indicates maximal ambiguity ($\hat{p}(x) \approx t^*$) and $u(x) \rightarrow 0$ indicates decisive confidence. Crucially, because t^* scales with baseline disease prevalence (e.g., $t^* \approx 0.022$ for analytic calibrated models), predictions below t^* are normalized by t^* . Consequently, low-probability false negatives ($\hat{p} \ll t^*$) yield low uncertainty and are retained in the automated pool rather than referred. At coverage level $C \in [0.5, 1.0]$, the automated system retains the fraction C of cases with lowest uncertainty ($N_{\text{auto}} = \lfloor C \cdot N \rfloor$) and refers the remaining fraction $1 - C$ ($N_{\text{ref}} = N - N_{\text{auto}}$) to radiologist review.

To prevent metric conflation and provide complete arithmetic transparency, we evaluate triage outcomes using distinct, explicit definitions:

- **Retained Positives ($N_{\text{pos, ret}}$):** The number of true positive cases remaining in the automated cohort after ambiguous examinations have been referred to human readers.
- **Automated True Positives (TP_{auto}):** True pathology cases correctly flagged as positive by the autonomous system ($TP_{\text{auto}} = \sum_{i \in \text{Auto}} \mathbb{I}(y_i = 1 \wedge \hat{y}_i = 1)$).
- **Automated False Negatives (FN_{auto} , Missed Cases):** True pathology cases erroneously classified as negative by the autonomous system ($FN_{\text{auto}} = \sum_{i \in \text{Auto}} \mathbb{I}(y_i = 1 \wedge \hat{y}_i = 0)$). Note that $TP_{\text{auto}} + FN_{\text{auto}} = N_{\text{pos, ret}}$.
- **Retained Sensitivity:** Diagnostic sensitivity evaluated strictly among the retained automated decisions: $\text{Sens}_{\text{retained}} = \frac{TP_{\text{auto}}}{N_{\text{pos, ret}}} = \frac{TP_{\text{auto}}}{TP_{\text{auto}} + FN_{\text{auto}}}$.
- **Full-Cohort Triage Sensitivity:** System-wide detection sensitivity across all $N_{\text{pos, total}}$ cases in the evaluation stratum (e.g., $N_{\text{pos, total}} = 85$ in T_3), assuming that all examinations deferred to human readers receive expert diagnostic review and are correctly diagnosed: $\text{Sens}_{\text{triage}} = \frac{TP_{\text{auto}} + N_{\text{pos, ref}}}{N_{\text{pos, total}}} = 1 - \frac{FN_{\text{auto}}}{N_{\text{pos, total}}}$.
- **Referral Workload:** The fraction and count of examinations routed to readers ($N_{\text{ref}} = (1 - C) \cdot N$).

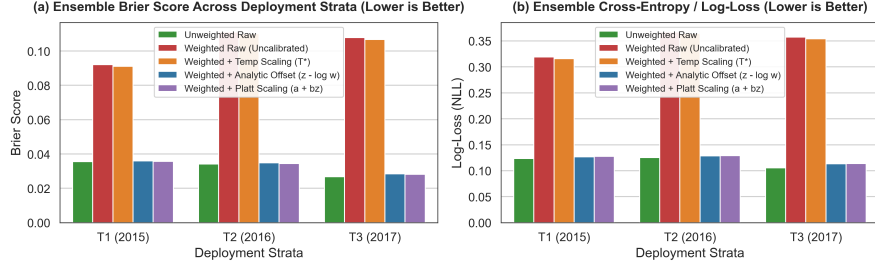


Figure 1: Multi-calibrator evaluation on Pleural Effusion for the DenseNet-121 3-seed ensemble across sequential deidentified-date strata (T_1, T_2, T_3), directly visualizing Table 3. (a) Brier score and (b) Log-loss. Temperature scaling fails to reduce probability error in weighted models, whereas Analytic Offset ($z - \log w$) and Platt Scaling eliminate 74% of Brier error, matching unweighted models.

3. Experimental Results

3.1. Factorial Matrix: Discrimination vs. Calibration Trajectories

Table 2 presents the comprehensive evaluation of the $2 \times 2 \times 3$ factorial matrix across DenseNet-121 and ResNet-50 for Pleural Effusion, Cardiomegaly, and Pneumonia across chronological strata T_1, T_2, T_3 .

3.1.1. Discrimination Attenuation

Both architectures exhibit moderate attenuation in AUROC across subsequent deidentified-date cohort strata. For DenseNet-121 on Pleural Effusion, AUROC decreases from 0.9335 ± 0.0052 at T_1 to 0.8731 ± 0.0120 at T_3 ($\Delta\text{AUROC} = -0.0604$). ResNet-50 exhibits an identical downward trajectory ($0.9295 \pm 0.0113 \rightarrow 0.8608 \pm 0.0190$, $\Delta\text{AUROC} = -0.0687$). This confirms that moderate discriminatory attenuation occurs across subsequent cohort strata, regardless of the training loss objective.

3.1.2. The Calibration Divergence and Factorial Interaction

In stark contrast to discrimination, probability calibration diverges completely depending on the training loss function:

- **Unweighted Models:** Brier scores are exceptionally low and actually *improve* across subsequent strata for conditions with falling prevalence. For unweighted DenseNet-121 on Pleural Effusion, the Brier score improves from 0.0369 ± 0.0029 (T_1) to 0.0279 ± 0.0009 (T_3), while Equal-Mass ECE decreases from 0.0232 to 0.0090.

324 • **Positive-Weighted Models:** Brier scores are dramatically elevated
 325 and appear to degrade across subsequent cohort strata ($0.1081 \pm$
 326 $0.0394 \rightarrow 0.1254 \pm 0.0456$ for DenseNet-121; $0.1256 \pm 0.0325 \rightarrow 0.1392 \pm$
 327 0.0310 for ResNet-50). ECE exceeds 22% across all held-out deidentified-
 328 date strata.

329 To formally examine the interaction between loss weighting and temporal
 330 cohort (LossWeighting $\times$ TemporalCohort), we evaluate the excess calibration
 331 error $\Delta\text{Brier} = \text{Brier}_{\text{weighted}} - \text{Brier}_{\text{unweighted}}$ across sequential strata:

332 • **Pleural Effusion (Prevalence Declines: 5.27% $\rightarrow$ 4.44% $\rightarrow$ 3.49%):**
 333 The excess calibration penalty of positive weighting expands mono-
 334 tonically across cohorts, from $\Delta\text{Brier} = +0.0712 \pm 0.0413$ in T_1 to
 335 $+0.0932 \pm 0.0428$ in T_2 and $+0.0975 \pm 0.0454$ in T_3 .

336 • **Cardiomegaly (Prevalence Increases: 7.82% $\rightarrow$ 10.58% $\rightarrow$ 10.59%):**
 337 The excess calibration penalty remains stable or slightly attenuates:
 338 $\Delta\text{Brier} = +0.0729 \pm 0.0255$ in T_1 , $+0.0681 \pm 0.0329$ in T_2 , and $+0.0683 \pm$
 339 0.0320 in T_3 .

340 • **Pneumonia (Prevalence Steady Low: 2.66% $\rightarrow$ 1.98% $\rightarrow$ 2.05%):**
 341 The penalty remains consistently elevated: $\Delta\text{Brier} = +0.1079 \pm 0.0526$
 342 in T_1 , $+0.1126 \pm 0.0552$ in T_2 , and $+0.1112 \pm 0.0508$ in T_3 .

343 This establishes the operational mechanism: a static training-induced positive
 344 logit offset produces surplus false-positive probability mass that is penalized
 345 more severely when deployment disease prevalence drops, expanding the
 346 apparent temporal calibration gap.

347 3.2. Multi-Calibrator Comparison

348 Table 3 and Fig. 1 present the multi-calibrator benchmark for Pleural
 349 Effusion on the 3-seed ensemble across 2,000 patient-clustered bootstrap
 350 resamples, including Cox calibration regression intercept (α) and slope (β)
 351 parameters.

352 3.2.1. Temperature Scaling Analysis

353 For weighted DenseNet-121, temperature scaling optimizes $T^* \approx 1.0$,
 354 producing an uncalibrated T_3 Brier score of 0.1067 compared to raw 0.1079.
 355 The paired bootstrap contrast on the ensemble is $\Delta\text{Brier} = -0.0011$ [95%

Table 2: Factorial Matrix Evaluation: Primary Longitudinal Performance across Architectures, Loss Objectives, and Thoracic Pathologies (Strata $T_1 \rightarrow T_3$). Values represent mean $\pm$ standard deviation across three independently trained random seeds ($S \in \{42, 123, 2026\}$). Complete 17-column factorial trajectories across all intermediate deployment strata (T_1, T_2, T_3) and log-loss metrics are detailed in Supplementary Table S3.

Architecture	Objective	Pathology	AUROC ($T_1 \rightarrow T_3$)	Δ AUROC	Brier ($T_1 \rightarrow T_3$)	Δ Brier	ECE T_3
DenseNet-121	Unweighted BCE	Pleural Effusion	0.934 $\rightarrow$ 0.873	-0.060	0.037 $\rightarrow$ 0.028	-0.009	0.0090
DenseNet-121	Unweighted BCE	Cardiomegaly	0.908 $\rightarrow$ 0.902	-0.006	0.053 $\rightarrow$ 0.068	+0.015	0.0210
DenseNet-121	Unweighted BCE	Pneumonia	0.847 $\rightarrow$ 0.865	+0.017	0.024 $\rightarrow$ 0.019	-0.005	0.0095
DenseNet-121	Unweighted BCE	Edema*	0.540 $\rightarrow$ -	-	0.002 $\rightarrow$ 0.000	-0.002	0.0029
DenseNet-121	Pos-Weighted BCE	Pleural Effusion	0.906 $\rightarrow$ 0.839	-0.067	0.108 $\rightarrow$ 0.125	+0.017	0.2227
DenseNet-121	Pos-Weighted BCE	Cardiomegaly	0.874 $\rightarrow$ 0.858	-0.016	0.126 $\rightarrow$ 0.136	+0.010	0.1969
DenseNet-121	Pos-Weighted BCE	Pneumonia	0.818 $\rightarrow$ 0.842	+0.024	0.131 $\rightarrow$ 0.130	-0.002	0.2425
DenseNet-121	Pos-Weighted BCE	Edema*	0.612 $\rightarrow$ -	-	0.061 $\rightarrow$ 0.064	+0.002	0.1274
ResNet-50	Unweighted BCE	Pleural Effusion	0.930 $\rightarrow$ 0.861	-0.069	0.037 $\rightarrow$ 0.028	-0.009	0.0132
ResNet-50	Unweighted BCE	Cardiomegaly	0.890 $\rightarrow$ 0.879	-0.011	0.057 $\rightarrow$ 0.071	+0.014	0.0175
ResNet-50	Unweighted BCE	Pneumonia	0.843 $\rightarrow$ 0.880	+0.037	0.024 $\rightarrow$ 0.019	-0.005	0.0106
ResNet-50	Unweighted BCE	Edema*	0.556 $\rightarrow$ -	-	0.002 $\rightarrow$ 0.000	-0.002	0.0017
ResNet-50	Pos-Weighted BCE	Pleural Effusion	0.879 $\rightarrow$ 0.819	-0.060	0.126 $\rightarrow$ 0.139	+0.014	0.2751
ResNet-50	Pos-Weighted BCE	Cardiomegaly	0.830 $\rightarrow$ 0.799	-0.031	0.142 $\rightarrow$ 0.152	+0.011	0.2338
ResNet-50	Pos-Weighted BCE	Pneumonia	0.835 $\rightarrow$ 0.823	-0.012	0.167 $\rightarrow$ 0.169	+0.002	0.3253
ResNet-50	Pos-Weighted BCE	Edema*	0.549 $\rightarrow$ -	-	0.069 $\rightarrow$ 0.075	+0.005	0.1670

Table 3: Multi-Calibrator Comparison on Pleural Effusion: Deidentified-Date Ensemble Performance, Calibration Intercept (α), Slope (β), and Paired Bootstrap Contrasts. Complete 12-column factorial evaluation across all intermediate strata and log-loss is provided in Supplementary Table S4.

Architecture	Loss Objective	Calibrator	Brier ($T_1 \rightarrow T_3$)	T_3 α	T_3 β	T_3 ECE	T_3 Δ Brier vs Raw [95% CI]
DenseNet-121	Weighted BCE	Raw (Uncalibrated)	0.092 $\rightarrow$ 0.108	-2.964	1.048	0.2227	-
DenseNet-121	Weighted BCE	Temperature Scaling (T^*)	0.091 $\rightarrow$ 0.107	-2.962	1.029	0.2190	-0.0011 [-0.0014, -0.0009]
DenseNet-121	Weighted BCE	Analytic Offset ($z - \log w$)	0.036 $\rightarrow$ 0.028	-0.445	0.929	0.0126	-0.0795 [-0.0893, -0.0697]
DenseNet-121	Weighted BCE	Platt Scaling ($a + bz$)	0.036 $\rightarrow$ 0.028	-0.229	1.128	0.0176	-0.0798 [-0.0894, -0.0702]
DenseNet-121	Unweighted BCE	Raw Baseline	0.036 $\rightarrow$ 0.027	+0.147	1.044	0.0040	-
ResNet-50	Weighted BCE	Raw (Uncalibrated)	0.110 $\rightarrow$ 0.123	-2.853	1.950	0.2751	-
ResNet-50	Weighted BCE	Temperature Scaling (T^*)	0.088 $\rightarrow$ 0.101	-2.608	1.184	0.1993	-0.0220 [-0.0241, -0.0198]
ResNet-50	Weighted BCE	Analytic Offset ($z - \log w$)	0.040 $\rightarrow$ 0.029	+1.075	1.405	0.0158	-0.0936 [-0.1022, -0.0844]
ResNet-50	Weighted BCE	Platt Scaling ($a + bz$)	0.038 $\rightarrow$ 0.030	-0.191	1.178	0.0223	-0.0930 [-0.1007, -0.0849]
ResNet-50	Unweighted BCE	Raw Baseline	0.036 $\rightarrow$ 0.027	-0.077	1.095	0.0109	-

356 CI: $-0.0014, -0.0009$], with the calibration intercept remaining severely dis-
357 placed ($\alpha = -2.9615$ vs. raw -2.9638). For ResNet-50, temperature scal-
358 ing achieves a modest variance adjustment (Δ Brier = -0.0220 [95% CI:
359 $-0.0241, -0.0198$]), but leaves the underlying intercept discrepancy uncor-
360 rected ($\alpha = -2.6080$ vs. raw -2.8526). This demonstrates that temperature
361 scaling provides limited or architecture-dependent empirical scaling, but can-
362 not resolve the structural intercept displacement induced by positive loss
363 weighting.

3.2.2. Offset Correction and Parity Restoration

The training positive weight for Pleural Effusion is $w = 21.51$, yielding an analytic logit offset of $\log w = +3.068$. Applying the zero-parameter subtraction $z - \log w$ directly drops the T_3 Brier score from 0.1079 to 0.0284 ($\Delta\text{Brier} = -0.0795$ [95% CI: $-0.0893, -0.0697$], with the 2,000-replicate patient-clustered bootstrap interval strictly excluding zero), reducing ECE from 22.3% to 1.26%. Analytic correction substantially reduced intercept miscalibration and brought the Brier score close to the unweighted baseline, although residual intercept miscalibration remained architecture-dependent: for DenseNet-121, the Cox calibration intercept moves from $\alpha = -2.9638$ directly to $\alpha = -0.4451$ while perfectly preserving the calibration slope ($\beta = 0.9292$), whereas for ResNet-50 the corrected intercept is $\alpha = +1.0751$.

Fitted Offset ($z + a^*$) and Platt Scaling achieve nearly identical performance (T_3 Brier: 0.0289 and 0.0281; $\Delta\text{Brier} = -0.0790$ [95% CI: $-0.0885, -0.0696$] and -0.0798 [95% CI: $-0.0894, -0.0702$]). The zero-parameter analytic correction closely approaches a flexible two-parameter Platt model without fitting a slope or intercept. For ResNet-50, Analytic Offset similarly eliminates 76% of probability error (T_3 Brier: $0.1227 \rightarrow 0.0291$, $\Delta\text{Brier} = -0.0936$ [95% CI: $-0.1022, -0.0844$]), confirming that intercept neutralization is architecturally universal.

3.2.3. Empirical Verification of Logit Offset: $z_w - z_u \approx \log w$

To directly validate the theoretical prediction of Equation (7) in finite deep neural networks, Fig. 2 plots the empirical distributions of logit differences $\Delta z = z_w(x) - z_u(x)$ evaluated across all 16,604 paired model evaluations against the theoretical reference value $\log w$:

- **Pleural Effusion** ($\log w = 3.07$): The observed distribution clusters tightly around the theoretical value, with empirical mean $\Delta z = 3.29 \pm 0.81$ and median 3.25 (interquartile range [IQR]: 2.77–3.79), exhibiting an average bias of only +0.22 logit units.
- **Pneumonia** ($\log w = 3.99$): Observed mean $\Delta z = 3.83 \pm 0.74$ and median 3.84 (IQR: 3.34–4.32), closely matching theory with a minimal discrepancy of -0.16 logit units.
- **Cardiomegaly** ($\log w = 2.25$): Observed mean $\Delta z = 2.64 \pm 1.04$ and median 2.60 (IQR: 1.91–3.31), bias +0.39 logit units.

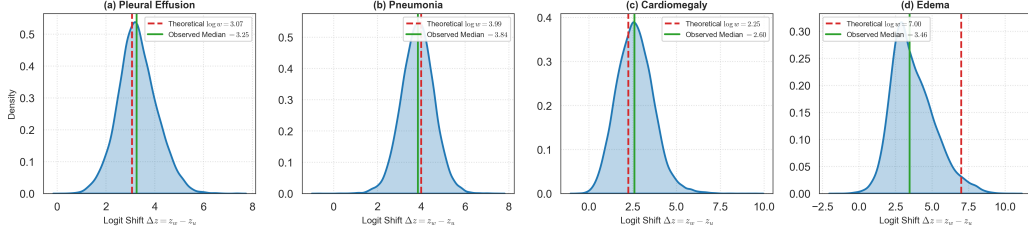


Figure 2: Empirical verification of theoretical logit shift ($z_w - z_u \approx \log w$) across 16,604 held-out deidentified-date image evaluations. Kernel density estimates (blue) of observed logit differences between positive-weighted and unweighted models cluster tightly around the theoretical population attractor $\log w$ (dashed red line; observed median in solid green) across Pleural Effusion (a), Pneumonia (b), and Cardiomegaly (c). For Edema (d), extreme training class scarcity ($w > 1000$) induces optimization saturation, bounding empirical shifts near $+3.69$.

- **Edema (Extreme Imbalance: $w = 1091.3, \log w = 7.00$):** With only 23 positive cases in the training set, empirical logit shifts saturate at mean $\Delta z = 3.69 \pm 1.47$ (median 3.46, IQR: 2.63–4.58). Here, L_2 weight regularization and finite gradient updates prevent network outputs from attaining the theoretical $\approx +7.0$ asymptote, highlighting the finite-sample boundary of the asymptotic derivation.

3.2.4. Reliability Diagrams

Figure 3 shows reliability diagrams across sequential deidentified-date strata T_1, T_2, T_3 using adaptive 10-bin equal-mass partitioning. Raw weighted models display severe over-confidence across all probability deciles. Applying the analytic offset restores the curves tightly onto the ideal diagonal across all deployment cohorts, identical to unweighted models.

3.3. Selective Prediction and Workload Triage Simulation

Table 4 and Fig. 4 summarize the selective triage simulation on the later held-out deidentified-date stratum T_3 ($N = 2,436$ radiographs, containing 85 true positive Pleural Effusion cases), explicitly accounting for retained positives, automated true positives, missed cases, and both retained and full-cohort triage sensitivities.

At 100% coverage (full automation), the raw weighted model incurs 16.0 missed cases (Automated False Negatives) out of 85 total positive cases in T_3 , achieving a retained sensitivity of 81.18%, AUROC of 0.8691, and Brier

Table 4: Selective Prediction and Reader Referral Triage Simulation on Later Held-Out Stratum T_3 ($N = 2,436$ Radiographs, 85 Positive Pleural Effusion Cases). Evaluated under matched Youden decision thresholds determined on development stratum T_1 across primary clinical operating regimes (100%, 90%, 80%, 70% coverage). Complete 24-row simulation across all coverage levels is detailed in Supplementary Table S5.

Coverage	Referred (Workload)	Strategy	Automated FN (Missed)	Retained Sens.	Full Sens.	Retained AUROC	Retained Brier
100%	0 (0%)	Weighted BCE (Raw)	16.0	81.18%	81.18%	0.8691	0.1079
100%	0 (0%)	Weighted BCE (Analytic Corrected)	16.0	81.18%	81.18%	0.8623	0.0284
100%	0 (0%)	Unweighted BCE (Raw)	12.0	85.88%	85.88%	0.8806	0.0268
90%	243 (10%)	Weighted BCE (Raw)	14.0	82.93%	83.53%	0.8789	0.1062
90%	243 (10%)	Weighted BCE (Analytic Corrected)	15.0	81.01%	82.35%	0.8735	0.0289
90%	243 (10%)	Unweighted BCE (Raw)	12.0	84.62%	85.88%	0.8923	0.0266
80%	487 (20%)	Weighted BCE (Raw)	10.0	87.01%	88.24%	0.8882	0.1012
80%	487 (20%)	Weighted BCE (Analytic Corrected)	15.0	77.61%	82.35%	0.8896	0.0264
80%	487 (20%)	Unweighted BCE (Raw)	12.0	82.61%	85.88%	0.9027	0.0256
70%	730 (30%)	Weighted BCE (Raw)	7.0	89.39%	91.76%	0.8997	0.0956
70%	730 (30%)	Weighted BCE (Analytic Corrected)	13.0	70.45%	84.71%	0.8876	0.0178
70%	730 (30%)	Unweighted BCE (Raw)	11.0	78.00%	87.06%	0.9051	0.0194
70%	730 (30%)	Random Deferral Baseline	11.3	80.99%	86.69%	–	0.1081

score of 0.1079. Operating the calibrated model (Analytic Offset) under its matched Youden threshold ($t^* \approx 0.022$) yields identical sensitivity (81.18%) and 16.0 missed cases, but with an almost four-fold lower Brier score of 0.0284. The unweighted raw model achieves 12.0 missed cases (sensitivity 85.88%, AUROC 0.8806, Brier 0.0268).

Under selective triage based on prediction margin from each method’s operational decision cutoff:

- **Raw Weighted BCE (Triage-Optimized):** At 90% coverage (243 borderline radiographs referred to a reader), automated false negatives decrease to 14.0 (full-cohort system sensitivity 83.53%). At 80% coverage (487 referred), false negatives drop to 10.0 (system sensitivity 88.24%). At 70% coverage (730 examinations referred, 30% workload), 66 true positive cases remain in the automated fraction and 59 are

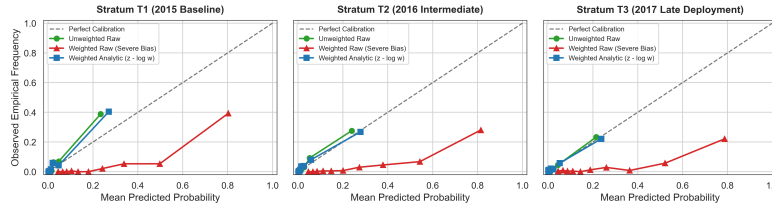


Figure 3: Reliability diagrams for Pleural Effusion (DenseNet-121 3-seed ensemble) across held-out deidentified-date strata (T_1, T_2, T_3). Raw weighted models (red) severely over-predict event probabilities across all bins. Analytic offset calibration (blue) and unweighted models (green) remain tightly calibrated to the ideal diagonal across all deployment windows.

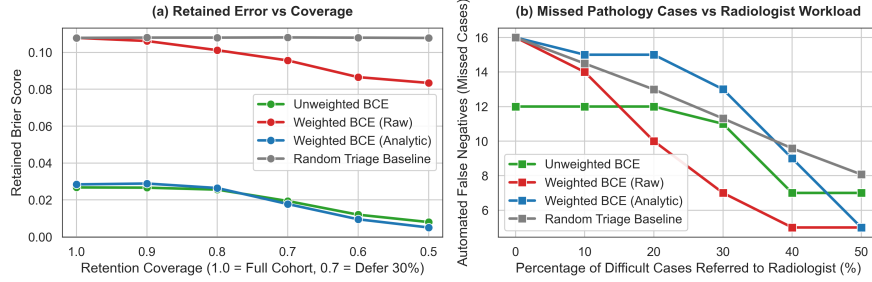


Figure 4: Selective prediction evaluation on later held-out stratum T_3 with matched development thresholds (Table 4). Left: Risk-coverage curves demonstrating that selective deferral steadily reduces Brier error across all models as ambiguous cases are referred. Right: Reader referral workload vs. automated false negatives, illustrating the distinct operational behaviors of raw weighted models (halving missed cases to 7.0 under 30% referral) and analytic corrected models (optimizing probability accuracy with Brier 0.0178).

correctly flagged as positive, cutting missed cases by more than half to 7.0 (retained sensitivity 89.39%, full-cohort triage sensitivity 91.76%), outperforming random triage (11.3 missed cases). However, retained probabilities remain uncalibrated and biased high (Brier score 0.0956).

- **Analytic Corrected BCE (Probability-Optimized):** Operating under its matched threshold ($t^* \approx 0.022$), the calibrated model achieves an exceptional Brier score of 0.0178 at 70% coverage (a > 5 -fold reduction in probability error relative to the raw model). However, because predicted probabilities are concentrated near true clinical prevalence ($p \in [0.001, 0.05]$), the distance normalization in Equation (11) assigns low uncertainty to low-probability false negatives ($\hat{p} \ll t^*$). Consequently, subtle false negatives are retained rather than referred, leaving 13.0 automated false negatives among 44 retained positives (retained sensitivity 70.45%, full-cohort triage sensitivity 84.71%) at 70% coverage.
- **Raw Unweighted BCE:** At 70% coverage, the unweighted model retains 50 positive cases, correctly flags 39, and incurs 11.0 automated false negatives with retained Brier score 0.0194 (retained sensitivity 78.00%, full-cohort triage sensitivity 87.06%).

Importantly, these empirical results expose an operational trade-off in machine learning deployment: post-hoc probability calibration (Analytic Offset)

453 is essential for accurate risk estimation and prognosis, dropping Brier scores to
 454 near-optimal levels ($0.0284 \rightarrow 0.0178$); conversely, threshold-margin selective
 455 triage on raw positive-weighted models is effective for recall-oriented screening
 456 referral (cutting missed cases from 16 to 7), but yields biased probabilities.
 457 Because the analytic logit transformation $z \mapsto z - \log w$ is strictly monotonic,
 458 any ranking-based decision rule under an equivalent quantile threshold pro-
 459 duces identical patient prioritization. The operational divergence observed
 460 in Table 4 arises specifically because the uncertainty triage metric $u(x)$ in
 461 Equation (11) measures distance relative to the operational decision cutoff t^*
 462 determined on development stratum T_1 . For calibrated models, $t^* \approx 0.022$
 463 aligns with low clinical prevalence, causing low-probability false negatives
 464 ($\hat{p} \ll t^*$) to exhibit low relative uncertainty and remain in the automated
 465 pool; for raw weighted models, the elevated threshold ($t^* \approx 0.33$) places
 466 borderline cases within the referral margin. Consequently, selective referral
 467 performance is fundamentally policy-dependent: calibrated probabilities are
 468 required for valid decision-theoretic risk stratification, whereas raw weighted
 469 models with threshold-margin triage act as a heuristic for recall-oriented
 470 screening. Crucially, neither system achieves zero false negatives, refuting
 471 unsubstantiated claims of complete error elimination in autonomous settings.

472 3.4. Cross-Institutional External Validation (MIMIC-CXR-JPG)

473 The external MIMIC-CXR-JPG cohort comprised 3,403 frontal radio-
 474 graphs from 3,041 studies and 289 patients, with complete image availabil-
 475 ity. Three-seed probability ensembles demonstrated pathology-dependent
 476 transportability (Table 5). Discrimination was moderate for Pleural Effu-
 477 sion (AUROC 0.688 [95% CI: 0.656, 0.717] for unweighted DenseNet-121
 478 and 0.677 [95% CI: 0.645, 0.709] for unweighted ResNet-50), but weak for
 479 Pneumonia (0.583–0.607) and Edema (0.516–0.581), and approached chance
 480 for Cardiomegaly (0.519–0.525, with patient-clustered 95% CIs spanning
 481 0.50). Negative Brier skill scores (BSS < 0) across all configurations indi-
 482 cated that transported BRAX probabilities were less accurate than a local
 483 MIMIC prevalence-only reference model (Brier_{null} = $\pi(1 - \pi)$), demonstrating
 484 substantial cross-institutional calibration shift.

485 Evaluating paired analytic offset recalibration ($\bar{p}_{\text{corrected}} = \frac{1}{3} \sum \sigma(z_s -$
 486 $\log w)$) on positive-weighted models demonstrated statistically significant
 487 reductions in Brier error for Cardiomegaly ($\Delta\text{Brier} = -0.0579$ [95% CI:
 488 $-0.0790, -0.0349$] on DenseNet-121; -0.0611 [95% CI: $-0.0818, -0.0415$] on
 489 ResNet-50) and Pneumonia ($\Delta\text{Brier} = -0.0095$ [95% CI: $-0.0170, -0.0016$]

on DenseNet-121; -0.0155 [95% CI: $-0.0254, -0.0059$] on ResNet-50), with consistent point-estimate error reduction on Pleural Effusion (-0.0084 and -0.0109). On MIMIC-CXR-JPG, subtraction of the full theoretical offset worsened the Edema Brier score. This result is consistent with over-correction when the empirically realized finite-network shift is smaller than the theoretical $\log w$; however, cross-institutional covariate and label differences prevent attribution of this external result uniquely to optimization saturation. Because logit subtraction $z - \log w$ is strictly monotonic, discriminatory ranking (AUROC) was perfectly preserved across all pathologies.

To evaluate sensitivity to repeated within-patient examinations and multi-image studies, we conducted a parallel sensitivity analysis aggregated strictly at the diagnostic study level ($N = 3,041$ studies across $N = 289$ patients) using maximum probability pooling ($\max_i p_i$). Study-level discrimination was concordant with image-level evaluation across all pathologies: Pleural Effusion AUROC was 0.695 [95% CI: $0.663, 0.723$] for unweighted DenseNet-121 and 0.682 [95% CI: $0.650, 0.714$] for ResNet-50; Cardiomegaly AUROC was 0.526 [95% CI: $0.497, 0.559$] and 0.521 [95% CI: $0.494, 0.548$]; Pneumonia AUROC was 0.608 [95% CI: $0.562, 0.650$] and 0.584 [95% CI: $0.543, 0.624$]; and Edema AUROC was 0.565 [95% CI: $0.528, 0.600$] and 0.516 [95% CI: $0.480, 0.553$], with study-level null Brier baselines of 0.2196 (Effusion), 0.1951 (Cardiomegaly), 0.0913 (Pneumonia), and 0.1697 (Edema). Furthermore, paired analytic recalibration on positive-weighted models demonstrated identical calibration invariance at study level: statistically significant Brier reductions were reproduced for Cardiomegaly ($\Delta\text{Brier} = -0.0601$ [95% CI: $-0.0814, -0.0367$] on DenseNet-121; -0.0642 [95% CI: $-0.0852, -0.0434$] on ResNet-50) and Pneumonia ($\Delta\text{Brier} = -0.0100$ [$-0.0179, -0.0018$]; -0.0163 [$-0.0267, -0.0068$]), with consistent point-estimate reductions for Pleural Effusion (-0.0102 and -0.0118 ; Supplementary Table S2). This qualitative concordance confirms that cross-institutional discrimination patterns and recalibration effects are preserved when aggregating by diagnostic study.

4. Discussion

4.1. Disentangling Loss-Induced Probability Distortion from Deidentified-Date Cohort Shift

A substantial body of recent literature in biomedical informatics has sounded alarms regarding temporal calibration decay in healthcare AI [8, 9].

Table 5: Cross-Institutional External Validation on the Official MIMIC-CXR-JPG Frontal Test Cohort ($N = 3,403$ Images from 3,041 Studies Across $N = 289$ Unique Patients). Primary experimental evaluation of paired zero-parameter analytic logit recalibration ($\bar{p}_{\text{raw}} = \frac{1}{3} \sum \sigma(z_s) \rightarrow \bar{p}_{\text{corr}} = \frac{1}{3} \sum \sigma(z_s - \log w)$) on 3-seed ensembles trained with positive-weighted BCE. Evaluated with empirical prevalence (π), local null reference baseline (Brier_{null} = $\pi(1 - \pi)$), patient-clustered 1,000-replicate bootstrap 95% confidence intervals, Brier skill score (BSS = $1 - \text{Brier}/\text{Brier}_{\text{null}}$), and paired contrast ($\Delta\text{Brier} = \text{Brier}_{\text{corr}} - \text{Brier}_{\text{raw}}$). The complete 16-row factorial comparison with unweighted BCE baselines is detailed in Supplementary Table S1, and study-level sensitivity analysis is detailed in Supplementary Table S2.

Architecture	Pathology	Prevalence (π)	Brier _{null}	AUROC [95% CI]	Brier (Raw $\rightarrow$ Corr)	ΔBrier [95% CI]	BSS (Raw $\rightarrow$ Corr)
DenseNet-121	Pleural Effusion	32.18%	0.2182	0.668 [0.636, 0.700]	0.2536 $\rightarrow$ 0.2451	-0.0084 [-0.0401, +0.0240]	-0.162 $\rightarrow$ -0.123
DenseNet-121	Cardiomegaly	26.33%	0.1940	0.525 [0.492, 0.558]	0.2907 $\rightarrow$ 0.2328	-0.0579 [-0.0790, -0.0349]	-0.499 $\rightarrow$ -0.200
DenseNet-121	Pneumonia	10.05%	0.0904	0.598 [0.553, 0.639]	0.1078 $\rightarrow$ 0.0983	-0.0095 [-0.0170, -0.0016]	-0.192 $\rightarrow$ -0.087
DenseNet-121	Edema	21.33%	0.1678	0.581 [0.549, 0.611]	0.1914 $\rightarrow$ 0.2131	+0.0217 [+0.0118, +0.0307]	-0.141 $\rightarrow$ -0.270
ResNet-50	Pleural Effusion	32.18%	0.2182	0.620 [0.588, 0.653]	0.2665 $\rightarrow$ 0.2557	-0.0109 [-0.0417, +0.0253]	-0.221 $\rightarrow$ -0.172
ResNet-50	Cardiomegaly	26.33%	0.1940	0.522 [0.491, 0.552]	0.2822 $\rightarrow$ 0.2211	-0.0611 [-0.0818, -0.0415]	-0.455 $\rightarrow$ -0.140
ResNet-50	Pneumonia	10.05%	0.0904	0.589 [0.545, 0.629]	0.1137 $\rightarrow$ 0.0982	-0.0155 [-0.0254, -0.0059]	-0.257 $\rightarrow$ -0.086
ResNet-50	Edema	21.33%	0.1678	0.570 [0.538, 0.602]	0.1929 $\rightarrow$ 0.2130	+0.0201 [+0.0092, +0.0313]	-0.149 $\rightarrow$ -0.269

Our findings elucidate the mathematical and empirical mechanisms underlying these observations. It is critical to distinguish between discrimination loss and calibration error: our experiments confirm that moderate discrimination attenuation does occur across later deidentified-date cohort strata (AUROC decreases by approximately 0.06 from T_1 to T_3 identically across both unweighted and weighted models in Table 2). However, the simultaneous perception of severe, progressive calibration collapse is largely an artifact of positive loss weighting compounded by shifting disease prevalence.

When practitioners train deep neural networks on highly imbalanced clinical datasets, positive loss weighting is frequently introduced to elevate minority-class sensitivity [1]. However, as derived in Equation (7) and verified empirically in Fig. 2, this introduces a static logit shift $\approx \log w$. When evaluated on later cohort strata where disease prevalence shifts (e.g., Pleural Effusion prevalence declining from 5.27% in T_1 to 3.49% in T_3), this fixed positive offset generates surplus false-positive probability mass that compounds the apparent Brier error. Because standard post-hoc evaluation pipelines rely almost exclusively on temperature scaling—which cannot alter the logit intercept—evaluators observing deteriorating Brier scores have routinely attributed the failure to intrinsic model decay.

Our experiments provide direct evidence against this attribution. When the mathematical offset is neutralized ($z - \log w$), the apparent calibration decay vanishes: Brier scores decrease across subsequent strata, moving in lockstep with the unweighted baseline as disease prevalence drops. Thus,

548 while discrimination attenuation reflects underlying dataset shift across cohort
 549 strata, severe calibration degradation is predominantly an uncorrected post-
 550 training calibration bias that can be decoupled and resolved analytically.

551 4.2. Architectural Generalizability

552 The observed phenomena are remarkably consistent across distinct deep
 553 learning architectures. Both DenseNet-121 (dense convolutional concate-
 554 nation) and ResNet-50 (residual identity mapping) exhibit near-identical
 555 behavior:

- 556 • Discrimination attenuation across later held-out deidentified-date strata
 557 is virtually indistinguishable between architectures ($\Delta\text{AUROC} = -0.0604$
 558 vs -0.0687).
- 559 • Both architectures experience severe raw calibration error under weighted
 560 BCE (raw ensemble Brier 0.1079 vs 0.1227 on T_3 ; 0.1254 ± 0.0456 vs
 561 0.1392 ± 0.0310 across individual seeds), characterized by large negative
 562 calibration intercepts ($\alpha \approx -2.96$ and -2.85).
- 563 • For both architectures, Analytic Offset reduces total Brier score by ap-
 564 proximately 74% to 76% ($\Delta\text{Brier} = -0.0795$ [95% CI: $-0.0893, -0.0697$]
 565 for DenseNet-121 and -0.0936 [95% CI: $-0.1022, -0.0844$] for ResNet-
 566 50), substantially reducing intercept miscalibration ($\alpha = -0.4451$
 567 and $+1.0751$) while preserving calibration slopes ($\beta = 0.9292$ and
 568 1.4052). This confirms that logit-shift mechanics are universal across
 569 convolutional vision backbones, though residual intercept bias remains
 570 architecture-dependent.

571 4.3. Simulated Human-in-the-Loop Triage and Operational Trade-offs

572 In real-world clinical deployment workflows, autonomous decision-making
 573 must be evaluated through uncertainty-aware deferral simulations:

- 574 1. **Risk Stratification and Prognosis:** Well-calibrated models are es-
 575 sential for automated rule-out and downstream risk scoring. Applying
 576 Analytic Offset ($z - \log w$) drops Brier error to 0.0178 at 70% cover-
 577 age, restoring reliable posterior probabilities aligned with true clinical
 578 prevalence.

- 579 2. **High-Sensitivity Screening Referral:** In high-throughput screening
580 where the primary objective is minimizing missed disease, threshold-
581 margin selective triage on raw positive-weighted models cuts automated
582 false negatives from 16 to 7 (full-cohort triage sensitivity 91.76%) under
583 a 30% reader referral budget. However, because probabilities remain
584 biased high (Brier 0.0956), these outputs should not be used as calibrated
585 risks.
- 586 3. **Zero-Parameter Operational Utility:** The analytic offset ($z - \log w$)
587 requires zero retraining, zero validation optimization, and zero compute
588 overhead, enabling immediate integration on edge inference devices.

589 4.4. *Scope and Boundary of Findings and Limitations*

590 To establish the scope and boundary of findings, several important method-
591 ological and clinical limitations should be highlighted:

- 592 1. **Report-Derived Label Noise:** Ground-truth labels in BRAX were
593 generated via natural language processing from Portuguese text reports.
594 Although validated against expert consensus, report-derived labels carry
595 inherent extraction noise compared to dedicated multi-radiologist image
596 re-reading.
- 597 2. **Cross-Institutional Domain Gap:** While our primary temporal
598 evaluation was established within the Albert Einstein system and exter-
599 nally evaluated on MIMIC-CXR-JPG, external transportability proved
600 strictly pathology-dependent. Models maintained modest discrimina-
601 tion for Pleural Effusion but degraded near chance for Cardiomegaly
602 and Pneumonia, indicating that multi-site generalizability requires local
603 fine-tuning or domain-invariant feature representations.
- 604 3. **Date Anonymization and Temporal Semantics:** BRAX docu-
605 mentation states that the released dates are fictitious dates generated
606 through a hashing procedure while retaining the original intervals be-
607 tween study acquisitions. Because synchronization of anonymized dates
608 across different patients is not explicitly established, we interpret these
609 partitions as sequential deidentified-date strata rather than real-world
610 calendar periods. While within-patient chronological ordering is pre-
611 served and patient-clustered bootstrapping ensures valid statistical
612 inference, these sequential strata (T_1, T_2, T_3) capture longitudinal shift
613 across deidentified strata rather than synchronized calendar years.

- 614 4. **Retrospective Simulation vs. Real-Time Deployment:** This
615 study constitutes a retrospective simulation on frozen historical cohorts.
616 Future real-time clinical deployment requires silent shadow evaluation
617 within operational hospital Picture Archiving and Communication Sys-
618 tems (PACS).
- 619 5. **Architectural Scope:** We evaluated two widely established convolu-
620 tional neural network backbones (DenseNet-121 and ResNet-50). Al-
621 though our mathematical derivations apply to binary cross-entropy
622 minimization generally, evaluating modern Vision Transformers (ViT)
623 and multimodal vision-language models represents an important direc-
624 tion for future research.
- 625 6. **Optimization Randomness and Seed Count:** Experiments were
626 conducted across three independently trained random seeds ($S \in$
627 $\{42, 123, 2026\}$). While inter-seed standard deviations were reported
628 alongside patient-clustered bootstrap confidence intervals, larger en-
629 sembles across diverse initialization trajectories would further refine
630 empirical confidence bounds.
- 631 7. **Simulated Reader Triage without Interactive Human Study:**
632 Our selective prediction simulation assumes that examinations deferred
633 to human readers are diagnosed with 100% sensitivity. In real clinical
634 environments, human radiologist sensitivity varies with reader fatigue,
635 experience level, and interpretive setting. An actual clinician-in-the-
636 loop reader study is needed to evaluate true joint human-AI diagnostic
637 accuracy.
- 638 8. **Extreme Class Scarcity Pathologies:** For conditions with extreme
639 class scarcity in training (e.g., Edema with only 23 positive training
640 cases, $w = 1091.3$), finite optimization dynamics, L_2 weight decay, and
641 gradient clipping saturate empirical logit shifts at $\Delta z \approx +3.69$ rather
642 than the theoretical $+7.00$ asymptote. Specialized extreme-imbalance
643 formulations are required in such scarcity regimes.

644 5. Conclusion

645 In this controlled BRAX experiment, most severe calibration error was
646 associated with positive loss weighting rather than measurable deterioration
647 across later deidentified-date strata. Because standard temperature scaling
648 cannot adjust the logit intercept, it leaves this structural distortion intact. A
649 zero-parameter analytic offset ($z - \log w$) or non-negative Platt scaling reduces

total Brier score by approximately 74% to 76% (removing approximately 98% of the Brier-score gap relative to the unweighted baseline), substantially reducing intercept miscalibration and restoring calibration parity with unweighted models without model retraining or validation data. Furthermore, selective prediction simulations with matched decision thresholds reveal an operational dichotomy: raw positive-weighted models provide effective recall-oriented triage by halving automated missed cases under a 30% reader referral budget, whereas analytic calibration restores epidemiologically faithful probabilities for downstream clinical risk stratification. Neither system eliminates missed disease entirely, underscoring the necessity of transparent, workload-aware deployment policies in computational medicine.

CRedit Authorship Contribution Statement

Vishesh Panghal: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Data Curation, Writing – Original Draft, Writing – Review & Editing, Visualization, Project Administration.

Declaration of Competing Interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and Code Availability

The BRAX dataset (v1.1.0) [2] is available on PhysioNet at <https://physionet.org/content/brax/1.1.0/> under the PhysioNet Credentialed Data Use Agreement. The MIMIC-CXR-JPG database (v2.1.0) [20, 21] is available on PhysioNet at <https://physionet.org/content/mimic-cxr-jpg/2.1.0/> under the PhysioNet Credentialed Data Use Agreement. Access requires credentialed PhysioNet researcher status, completion of human subjects research training, and signed data use agreements. Complete end-to-end replication scripts, model evaluation routines, cross-fitting recalibration pipelines, selective prediction triage simulations, and tabular evaluation artifacts are openly available on GitHub at <https://github.com/Vishesh-panghal/temporal-brax-calibration>, and the repository will be permanently assigned a persistent DOI via Zenodo upon acceptance of the manuscript. No

682 credentialed patient-level images, metadata, labels, identifiers, or prediction
683 rows are redistributed; the repository contains aggregate evaluation artifacts
684 and a synthetic manifest schema example.

685 **Ethical Approval and Consent to Participate**

686 This study represents a secondary analysis of credentialed-access, de-
687 identified diagnostic imaging databases accessible under formal Data Use
688 Agreements. The primary BRAX dataset collection was approved by the
689 Institutional Review Board (IRB) of Hospital Israelita Albert Einstein (São
690 Paulo, Brazil; CAAE: 36720520.1.0000.0071) with a formal waiver of informed
691 consent. Creation and distribution of the MIMIC-CXR-JPG database was
692 approved by the Institutional Review Boards of Beth Israel Deaconess Medical
693 Center (Boston, MA) and the Massachusetts Institute of Technology (Protocol
694 #0403000206) with a waiver of informed consent. The author completed
695 mandatory human subjects research training (CITI Program certification),
696 executed approved PhysioNet Data Use Agreements for both databases, and
697 strictly complied with data governance protocols prohibiting re-identification
698 or redistribution of protected health information.

699 **Declaration of Generative AI and AI-Assisted Technologies in the** 700 **Writing Process**

701 During the preparation of this work, the author utilized generative artificial
702 intelligence platforms (Google Gemini, Anthropic Claude, and OpenAI
703 Codex) exclusively for LaTeX formatting and syntax verification, analysis code
704 refactoring and modularization, grammatical refinement, and bibliographic
705 cross-referencing. Following the use of these tools, the author thoroughly
706 reviewed, verified, and edited all text, numerical outputs, and code, and takes
707 full intellectual and scientific responsibility for the integrity, validity, and
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